The Penrose Class: Personalised Version of the LaTeX Article Class

Documentation and Demonstration for `penrose.cls` Version 2.0

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1 Introduction

In this example document we describe how to use the `penrose` class. This is not a minimal working example, and might not cover all the features of this class. One should see the source code of this PDF for a more technical introduction. We have also used the `mpt-core` and `mpt-thm` packages written by the author to include brief samples of mathematical content. These are completely optional, and the user can use their own mathematical packages if needed.

2 Metadata and The Title Block

The class provides a flexible way to handle your document information. You can specify a subtitle and affiliation, which are automatically formatted in the title block. Note that the `\title` command accepts an optional argument in square brackets; this is what appears in the page headers from page two onwards, preventing long titles from overflowing the header space.

Similarly, the `\date` command is versatile. If you prefer a formal look, the standard command adds a "Date:" prefix. If you want a cleaner look, you can use the starred version `\date*\{...\}` to print the date on its own.

Use `\maketitle` to display the title block. Both the `\maketitle` command and the `abstract` environment are optional and should be used after `\begin{document}`.

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3 Tables and Columns

The class extends standard tables by providing custom column types through `tabularx` and `booktabs` packages. These are particularly useful when you need precise control over alignment and width. These are listed in table (1). Additionally the standard column type `l, r, c, p{<length>}, X` defined by `tabularx` are also available.

Table 1 – Custom column types

Type	Description
`C`	Type `X` with centre aligned text
`L`	Type `X` with left aligned text
`R`	Type `X` with right aligned text
`D{len}`	Type `p{len}` with centre aligned text
`U{len}`	Type `p{len}` with left aligned text
`V{len}`	Type `p{len}` with right aligned text

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4 Section Heading

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4.1 Subsection Heading

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Theorem 4.1 (Global Hyperbolicity)**.** *A spacetime* (M, g) *is globally hyperbolic if it contains a Cauchy surface.*

Equation (1) is known as the Einstein's field equation of General Relativity.

$$R_{ab} - \frac{1}{2} R g_{ab} = 8\pi G T_{ab} \tag{1}$$

For citations, the class is pre-configured with `natbib` and uses the `naturemag` bibliography style. And finaly here are some example of citations [1], [2], [1, page -1].

$$a^2 + b^2 = c^2 \tag{2a}$$

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$$a^3 + b^3 = c^3 \tag{2b}$$

Equations (2a) and (2b) are subequations.

4.1.1 Subsubsection Heading

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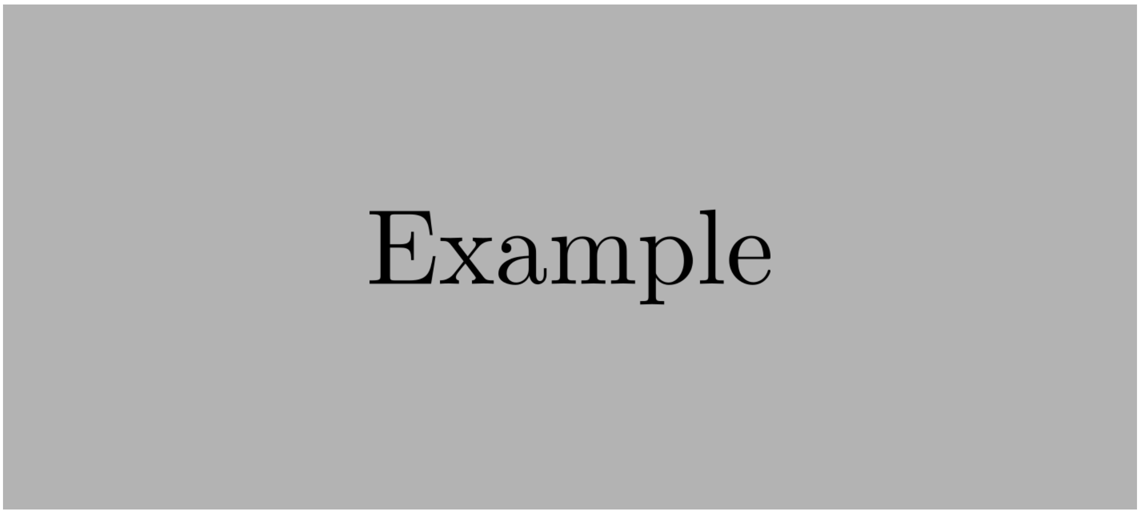

Figure 1 – A sample figure showing the clean captioning style of the class.

4.2 Subsection Heading

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5 Section Heading

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5.1 Subsection Heading

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In case you want a wider text block. You can use the `widetext` environment.

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In the other direction, there is a `quote` environment.

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6 Conclusion

After procrastinating for hours, I finally gave up on avoiding AI and asked ChatGPT to write a concluding paragraph. This is what it had to say: The `penrose` class is designed to be out-of-the-way. It handles the heavy lifting of page geometry and headers so you can focus on the Lie algebras or the gravitational flux. By keeping the design minimal, it ensures that the document remains timeless.

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